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304 Stainless Steel

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EN 1.4301 grade 304 Stainless Steel 18/8 (18% chromium 8% nickel) is the most widely used of all Stainless Steel types. Due to its composition, corrosion and heat resistance and its mechanical properties, grade 304 is used in all fields of industrial, commercial and domestic manufacturing.

Corrosion and Oxidisation Resistance of 304

304 Stainless Steel shows excellent corrosion and oxidisation resistance in a range of atmospheric environments and many corrosive substances.

It is resistant to potable water containing up to approximately 200mg/L chlorides at room temperature 20°C (68°F) to 25°C (77°F), this reduces to about 150mg/L at around 60°C (140°F).

EN 1.4301 304 Stainless Steel provides good oxidisation resistance in intermittent service to 870°C (1,598°F) and continuous service to 925°C (1697°F) The use of 304 Stainless Steel is not recommended in the range of 425-860°C (797-1580°F) if subsequent aqueous corrosion resistance is required.

Chemical Composition of 304

	C	Mn	Si	P	S	Cr	Mo	Ni	N
Min						18.0		8.0	
Max	0.08	2.0	0.75	0.045	0.030	20.0		10.5	0.10

Welding

304 Stainless Steel provides excellent welding with and without filler metals using standard fusion methods. Post-weld annealing is not required when welding thin sections however heavy welded sections may require some post-weld annealing for optimal corrosion resistance.

For optimal annealing heat to 1010-1120°C (1850-2048°F) and cool rapidly with air or water.

EN 1.4301 304 Stainless Steel cannot be hardened by thermal treatment.

Mechanical Properties of 304

Yield Strength 0.2% (MPa) min	Tensile strength (MPa) min	Elongation (% in 50mm) min	Rockwell B (HR B) max	Brinell (HB) max
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205	515	40	92	201
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Physical Properties of 304

Density (kg/ m cube)	Elastic Modulus (GPa)	Specific heat 0-100°C (J/kg.K)	Electrical resistivity (nΩ.m)
8000	193	500	720

Mean Coefficient of Thermal Expansion (µm/m/°C)

0-100°C	0-315°C	0-538°C
17.2	18.4	16.2

Thermal Conductivity (W/m.K)

at 100°C	at 500°C
16.2	21.5



20 hours ago

Metals4U would use again

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Mr Chris Dunn



First c

First c
delive

Brian

For Trade & DIY

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